INSPECTION REPORT

Pressure Vessel V-301 — Propane Surge Drum — Statutory Inspection

Document ID	INS-2024-061	**Plant / Unit**	BHLV — Tank Farm West
Document Type	Statutory Inspection Report	**Date Issued**	05 June 2024
Revision	Final	**Classification**	INTERNAL — CONTROLLED

1. Equipment Summary

Equipment Tag	V-301
Description	Vertical Pressure Vessel — Propane Surge Drum
Design Code	IS 2825 / ASME Sec. VIII Div. 1
Design Pressure	24.0 bar g
Design Temperature	65 °C
Material	SA-516 Gr. 70 (carbon steel)
Year of Fabrication	2011
Last Inspection	May 2021 (INS-2021-022)
Current Inspection	03–04 June 2024
Inspector	K. Rajan + TPI: G. Iyer, Bureau Veritas India

2. Inspection Findings

2.1 Shell & Heads

Radiographic examination (RT) of longitudinal weld seam W-01 and circumferential weld seams W-02, W-03: no reportable indications. Visual inspection of external shell — minor atmospheric corrosion; surface preparation and repaint required. Insulation removed from lower shell quadrant for UT. Minimum wall thickness measured: 14.6 mm (nominal 16.0 mm; minimum allowable 11.2 mm). Corrosion rate extrapolated from 2021 data: 0.12 mm/year — within acceptable band.

2.2 Nozzles & Drain Valves

FINDING — Action Required: Drain valve DV-301-03 stem packing found in deteriorated condition — consistent with the observation noted in MNT-2024-089 (PMP-203 Maintenance Log,

February 2024) and directly implicated in the gas accumulation near-miss NMR-2024-012 (March 2024). Packing replacement confirmed completed by A. Nair (08 March 2024 per NMR-2024-012). Valve stem inspected at this inspection — packing appears newly installed and serviceable. No weeping observed. Valve body ultrasonic tested — no wall thinning.

All other nozzles (inlet N1, outlet N2, PSV connection N4, instrumentation N5/N6) — serviceable. PSV-301 pulled and bench-tested by A. Nair's team: set pressure confirmed 27.5 bar g (within +/- 3% of nameplate). PSV reinstated under PTW-2024-089.

2.3 Pressure Test

Test Type	Test Pressure	Duration	Result
Hydrostatic (shell)	36.0 bar g	30 minutes	PASS
Leak test (pneumatic)	5.0 bar g	15 minutes	PASS

3. Recommendations

1. V-301 cleared for continued service at design pressure 24.0 bar g and design temperature 65 °C.

2. External shell — surface preparation and recoating to be completed within 90 days. A. Nair responsible.

3. UT corrosion monitoring grid to be re-established at this outage; next UT survey due June 2025.

4. Next statutory inspection due: June 2027 (3-year interval, Factory Act).

5. Lessons from NMR-2023-008 and NMR-2024-012 should be incorporated into V-301 operating manual — particularly the risk of overpressure under VLV-507 manual override and still-air gas accumulation risk.

4. Authorisation

Role	Name	Signature	Date
Sr. Process Engineer	K. Rajan	____________	05 Jun 2024
TPI Inspector	G. Iyer (BV)	____________	05 Jun 2024
Maintenance Supervisor	A. Nair	____________	05 Jun 2024
Plant Manager	R. Krishnan	____________	06 Jun 2024

Cross-references: NMR-2023-008, NMR-2024-012, MNT-2024-089, SOP-001, INS-2024-047.